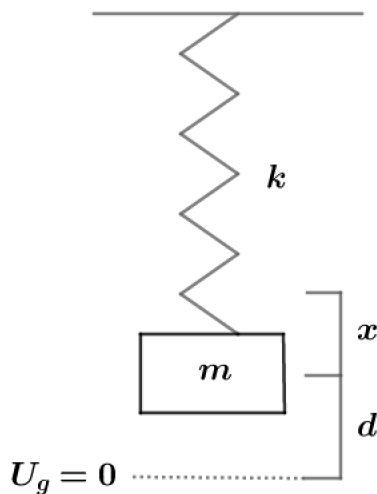


2011 F=ma Exam: Problem 15

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The initial energy of the spring-block system is

$$E_i = \frac{1}{2}k(x + d)^2$$

where x is the new equilibrium and d is the subsequent displacement. Note we have the initial position of the mass as $U_g = 0$. When the mass passes through the new equilibrium, the energy is

$$E_f = \frac{1}{2}kx^2 + mgd + \frac{1}{2}mv^2$$

From energy conservation,

$$\begin{aligned} \frac{1}{2}k(x + d)^2 &= \frac{1}{2}kx^2 + mgd + \frac{1}{2}mv^2 \\ kxd + \frac{1}{2}kd^2 &= mgd + \frac{1}{2}mv^2 \end{aligned}$$

Since $kx = mg$,

$$\frac{1}{2}kd^2 = \frac{1}{2}mv^2$$

We could have also gotten this directly using the equivalence of a vertical spring to a horizontal spring with shifted equilibrium. Thus,

$$k = \frac{mv^2}{d^2} = \frac{(0.100 \text{ kg})(0.75 \text{ m/s})^2}{(0.02 \text{ m})^2} = 140 \text{ N/m}$$

so the answer is D.